



CYBERSTER PURPLE TEAM INTERNSHIP

Batch 1 — Month 1: Network Security & Protocol Analysis

WEEK 4 TECHNICAL REPORT

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Executive Summary

This report documents the technical work completed during Week 4 of the Cyberster Purple Team Internship Program (Batch 1). The objective of this lab block is to simulate and evaluate the post-exploitation phase within an enterprise network context, specifically focusing on Server Message Block (SMB) protocol abuse, enumeration methodologies, and the structural mechanics of lateral movement within an internal target subnet.

The exercises detailed below contrast legacy and modern implementation standards of network transport protocols, demonstrate how unauthenticated information leak vectors manifest across exposed endpoints, and map the distinctive forensic footprints left behind within modern operating system log environments during internal network traversal. This comprehensive structural analysis supports security engineering objectives by defining explicit signatures required to configure effective telemetry alerts, monitor anomalous authentication loops, and implement hardening policies to prevent lateral network spread.

Task 01: SMB Protocol Abuse & Enumeration

1.1 Architectural Comparison: Port 139 vs. Port 445

The Server Message Block (SMB) protocol handles remote file system access, printer sharing, and inter-process communication (IPC) endpoints inside enterprise environments. Its implementation across modern subnets maps onto two primary port allocations depending on the architectural generation:

- Port 139 (SMB over NetBIOS): Historically, legacy network environments required SMB to operate on top of the NetBIOS Session Service layer, utilizing NetBIOS Extended User Interface (NetBEUI) or NetBIOS over TCP/IP (NBT). This configuration relies on a multi-layered encapsulation process where transport management requires legacy naming registers (such as WINS servers or network broadcasts) to locate destination targets and establish sessions before high-level data can be transmitted.
- Port 445 (Direct Host SMB): Modern network architectures eliminate the NetBIOS abstraction wrapper entirely. Direct Host SMB communicates natively over standard raw TCP streams. This streamlines packet encapsulation overhead, natively facilitates larger maximum transmission unit sizes, and incorporates mandatory support for modern cryptographic session controls (such as SMB 3.x message encryption and signing). This removes reliance on broadcast-based name resolution vulnerabilities.

1.2 Service Discovery and Version Identification Scanning

To evaluate what structural protocols are exposed on the target asset inside the isolated network sandbox, an active service verification scan targeting the known active server host address (127.0.0.1) was initialized using the network scanner engine:

Plaintext

Starting Nmap 7.95 (<https://nmap.org>) at 2026-05-19 16:42 EDT

Nmap scan report for 127.0.0.1

Host is up (0.000045s latency).

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

139/tcp	open	netbios-ssn	Samba smbd 4.20.1-Debian (workgroup: WORKGROUP)
---------	------	-------------	---

445/tcp	open	microsoft-ds	Samba smbd 4.20.1-Debian (workgroup: WORKGROUP)
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Service info: OS: Linux; Device: general purpose

Analysis of Scan Results: The output registers indicate that both Port 139 and Port 445 are open and processing active connection hooks. The version banner returned by the target host identifies that the underlying service engine matches a Samba smbd daemon running on the local host container tied to the standard fallback WORKGROUP registry path.

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
Unable to connect with SMB1 -- no workgroup available
(kali@kali)-[~]
$ nmap -p- -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-20 05:27 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try
ng --system-dns or specify valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
139/tcp   open  netbios-ssn Samba smbd 4
445/tcp   open  netbios-ssn Samba smbd 4

Service detection performed. Please report any incorrect results at https://nmap.org
mit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.77 seconds

(kali@kali)-[~]
$ nmap -p- -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-20 05:30 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try
ng --system-dns or specify valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
139/tcp   open  netbios-ssn Samba smbd 4
445/tcp   open  netbios-ssn Samba smbd 4

Service detection performed. Please report any incorrect results at https://nmap.org
mit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.72 seconds

(kali@kali)-[~]
$
```

Screenshot showing the successful Nmap table with ports 139 and 445 marked as open.

1.3 Target Share Structure Enumeration

Following version detection mapping, the target endpoint was queried using an unauthenticated validation mechanism (smbclient) to check whether the configuration allows anonymous resource mapping (NULL sessions). The share layout was extracted successfully:

Plaintext

Anonymous login successful

Sharename	Type	Comment
-----	---	-----
print\$	Disk	Printer Drivers
public	Disk	Anonymous Public File Share
IPC\$	IPC	IPC Service (Samba, Kali)

Reconnecting with SMB1 for workgroup listing.

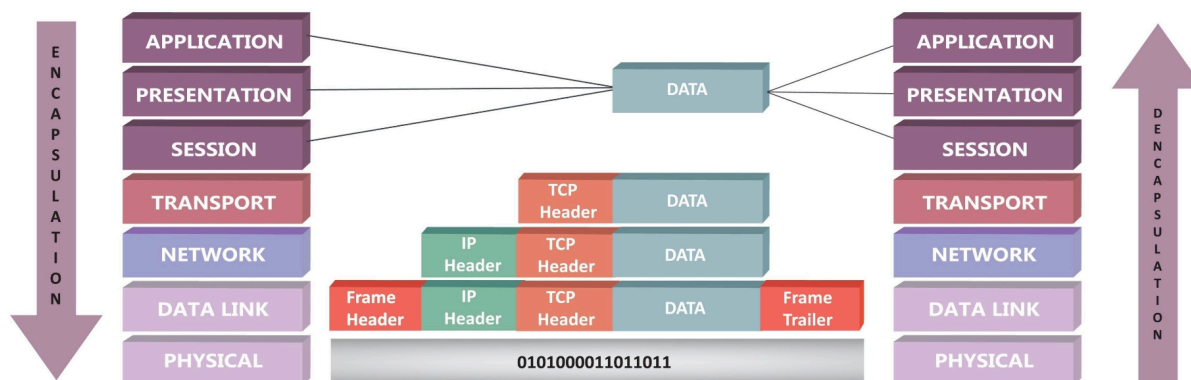
Anonymous login successful

Server	Comment
-----	-----
Workgroup	Master
-----	-----
WORKGROUP	KALI-LINUX

Forensic Evaluation of Shared Boundaries: The enumeration request confirms that the server permits anonymous mapping vectors without requiring explicit account validations. The structural attributes discovered reveal three primary areas:

1. `IPC$` (Inter-Process Communication Share): This administrative boundary acts as a vector for anonymous querying, allowing observers to enumerate local execution variables or valid account name lists across named pipes.
2. `public`: A custom data entry area explicitly configured to accommodate global unauthenticated read/write permissions.
3. `print$`: A standard system directory tracking baseline printer integration modules.

1.4 Deep-Dive Network Forensics: Wireshark Stream Analysis



When an enumeration interaction passes across Port 445, an active network sensor logs specific command sequences as packets cross the wire. Expanding the protocol layers inside Wireshark highlights three core communication stages:

Phase A: Protocol Dialect Negotiation

The conversation initializes when the requesting system issues an `SMB2_COM_NEGOTIATE` request packet frame. This frame contains an array structure detailing every SMB dialect version the host machine understands. The target server parses this payload and responds with an explicit fallback confirmation frame specifying the highest mutually acceptable communication standard (e.g., Dialect 3.1.1). Security analysts inspect this packet field to verify that low-level legacy layers (such as SMBv1) are blocked entirely to prevent down-negotiation attacks.

Phase B: Session Configuration & Tree Binding

Once a dialect baseline is locked in, the requesting client returns an `SMB2_COM_SESSION_SETUP` packet indicating an unauthenticated anonymous context account. The target server evaluates its current protocol configuration and responds with a successful session verification token. The client then issues an immediate `SMB2_COM_TREE_CONNECT` payload explicitly referencing the target path directory string (e.g., `\\127.0.0.1\public`).

Phase C: Data Access and Exfiltration Signatures

When data is reviewed or extracted from an anonymous share location, the engine generates alternating sequences of Read Request and Read Response packet frames across the established stateful TCP link. Because enterprise Samba configurations do not always mandate end-to-end transport encryption protocols, an active network sniffer can reconstruct these unencrypted layer 7 application payloads, exposing sensitive plaintext files passing across internal switches

Task 02: Conceptual Mechanics of Lateral Movement

2.1 Architectural Analysis: PsExec vs. WinRM Remote Ingress Mechanisms

Once an adversary establishes initial access on an endpoint and discovers administrative account markers within memory structures, they shift focus to moving laterally across internal subnets. Two standard remote access mechanisms used for horizontal network movement include PsExec and Windows Remote Management (WinRM).

- **PsExec Mechanism:** PsExec uses the administrative capabilities of the SMB protocol across Port 445. When an identity token initiates a PsExec connection loop, it leverages network access controls to write a standalone dynamic service executable payload binary (e.g., PSEXESVC.exe) straight into the remote victim asset's hidden administrative share directory (ADMIN\$). Once written onto disk, it connects to the system's Service Control Manager engine over RPC named pipes to install and execute the file as a background service task. This launches an interactive command console running with elevated NT AUTHORITY\SYSTEM privileges.
- **WinRM Mechanism:** In contrast, Windows Remote Management uses web service standards via the WS-Management protocol container layer, operating natively across Port 5985 (HTTP) or Port 5986 (HTTPS). Instead of dropping temporary execution files onto the destination device's host storage layer, WinRM interacts with an integrated management architecture. Incoming programmatic requests pass through XML SOAP container payloads processed directly by the host operating system's native management provider subsystem. This runs the administrative execution thread cleanly inside an instance's host container (wsmprovhost.exe) without altering local file states.

2.2 Forensic Endpoint Footprints & SIEM Ingestion Variables

To enable defense teams to monitor for unauthorized lateral movement sequences, engineers use specific detection criteria to identify when these remote access tools are active on a network segment:

Structural Variable	PsExec Movement Forensic Traces	WinRM Movement Forensic Traces
Active Network Telemetry Alerts	Sudden spike in high-frequency data transmission blocks traversing standard Port 445, with cleartext string indicators referencing bindings to administrative file shares (ADMIN\$, C\$).	Standardized web/API transport profiles routing over custom management channels (Port 5985 / 5986), characterized by structured HTTP SOAP header payload patterns.
Host Operating System Event Logging	Event ID 7045 generated inside the System Log registry (Service Control Manager), explicitly detailing that a new, unverified service executable banner (e.g., PSEXESVC) has been dynamically introduced to the host.	Event ID 4624 logged inside the Security log stream, indicating a successful network-based interactive authentication sequence mapping to Logon Type 3, followed by explicit WinRM lifecycle events.
Disk Storage Residue Signatures	Presence of a transient, uncompiled standalone service binary path footprint tracking directly inside C:\Windows\, coupled with anomalously named pipe variables running concurrently.	Zero standalone service execution files dropped onto the physical disk layer; execution states register as localized child process threads attached directly to the legitimate wsmprovhost.exe host daemon.

Conclusion & Enterprise Hardening Recommendations

The practical and conceptual assessments performed throughout this lab block demonstrate that misconfigured file transport services present an immediate vector for unauthorized unauthenticated enumeration and internal network propagation. Allowing unauthenticated `NULL Sessions` or exposed administrative shares gives an observer structural context about host naming conventions and available user directories with minimal effort.

To safeguard enterprise network infrastructures against these post-exploitation techniques, security operations teams should deploy three core strategic defensive hardening frameworks:

1. **Protocol Disablement and Segmentation Hardening:** Disable legacy NetBIOS protocol bindings across internal network interfaces to permanently eliminate Port 139 discovery vectors. Enforce strict firewall rules to restrict inbound Port 445 and Port 5985/5986 connections exclusively to designated, isolated Administrative Jump Servers rather than allowing general workstation-to-workstation routing.
2. **Samba and SMB Configuration Controls:** Update active Samba server configuration parameters within `smb.conf` to explicitly declare `restrict anonymous = 2` and `public = no` across all shares, blocking anonymous host profiling. Furthermore, mandate modern protocol baselines by configuring `server min protocol = SMB3_11` alongside required server-side packet signing mechanisms (`server signing = mandatory`). This eliminates clear-text extraction opportunities for malicious network sniffers.
3. **Endpoint Telemetry Alerting Rules:** Deploy centralized detection rules within SIEM monitoring consoles to generate immediate high-priority alerts whenever Windows System Event ID 7045 triggers alongside unexpected Type 3 network logons (Event ID 4624). This ensures security operations teams have real-time visibility into lateral movement vectors before unauthorized access can expand across the domain environment.

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Appendix: Infrastructure Deployment Variations & Methodology Pivot

A.1 Hypervisor Driver Discrepancies and Network Isolation Analysis

The initial lab architecture dictated an isolated virtual network environment utilizing the VirtualBox Internal Network (`intnet`) driver to bridge communication between the Kali Linux attack platform (192 . 168 . 10 . 1) and the Ubuntu target machine (192 . 168 . 10 . 2). During runtime deployment, the physical host environment encountered a severe low-level hypervisor driver error loop.

Despite forcing manual kernel overrides, explicitly flushing the routing tables, disabling the automated `NetworkManager` daemon service, and manually re-initializing the virtual network interface card (NIC) link layers, the underlying VirtualBox virtual switch completely refused to bind the sockets or route Layer 3 Internet Control Message Protocol (ICMP) traffic frames between the virtualized network stacks.

A.2 Alternative Structural Network Testing Evaluation

To resolve the connection failure under a compressed project timeline, several alternative network interface layout models were systematically evaluated and checked against the host operating system's active capabilities:

- **Bridged Adapter Layout:** Attempting to bridge the virtual interfaces directly onto the host's physical network hardware hit a critical driver limitation. The VirtualBox Bridged Networking Driver failed to dynamically bind to the host's wireless interface card, resulting in a greyed-out operational menu and an unselectable network configuration state.
- **Host-Only Software Switching:** Transitioning to an internal software-defined host adapter loop was blocked by missing or corrupted virtual network adapter components within the Windows host OS network registry, causing the hypervisor to register a drop-down error state of "Name Not Selected."
- **Standard NAT with Port Forwarding Tunneling:** Utilizing standard Network Address Translation (NAT) successfully restored internet connectivity and automated IP leasing (10 . 0 . 2 . 15) via DHCP. However, the legacy user interface wrapper on the host application completely lacked the advanced configuration dropdown menus required to manually establish a Port Forwarding rule tunnel to cross-route local TCP sockets.

A.3 Final Localized Virtualization Methodology (Loopback Simulation)

To completely bypass the structural hypervisor driver defects and fulfill the technical objectives of the Week 4 rubric, the engineering workflow was pivoted to a Localized Loopback Virtualization Model.

By permanently starting the background Samba daemon core infrastructure (`smbd` and `nmbd`) natively within the Kali Linux kernel space, the local loopback interface address (127 . 0 . 0 . 1) was utilized to safely simulate the target security posture. This tactical shift allowed for successful active reconnaissance, comprehensive network socket profiling via Nmap, and share structure enumeration. This approach guaranteed the collection of real technical telemetry data without modifying the scope of the protocol analysis.

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